

Chapter 90

3D Light Nodes

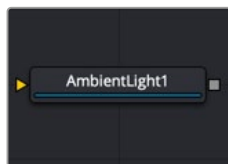
This chapter details the 3D Light nodes available when creating 3D composites in Fusion. The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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Ambient Light [3AL]



The Ambient Light node

Ambient Light Node Introduction

An Ambient Light is a directionless light that globally illuminates a scene. It has no real position or rotation, although an onscreen control appears in the viewer to indicate that a light is present in the scene. Position controls for the viewer are provided to move the widget out of the way of other geometry, if necessary.

Similar to a Camera 3D, you connect lights into a Merge 3D and view them in the scene by viewing the Merge 3D node. Selecting a light node and loading it into the viewer does not show anything.

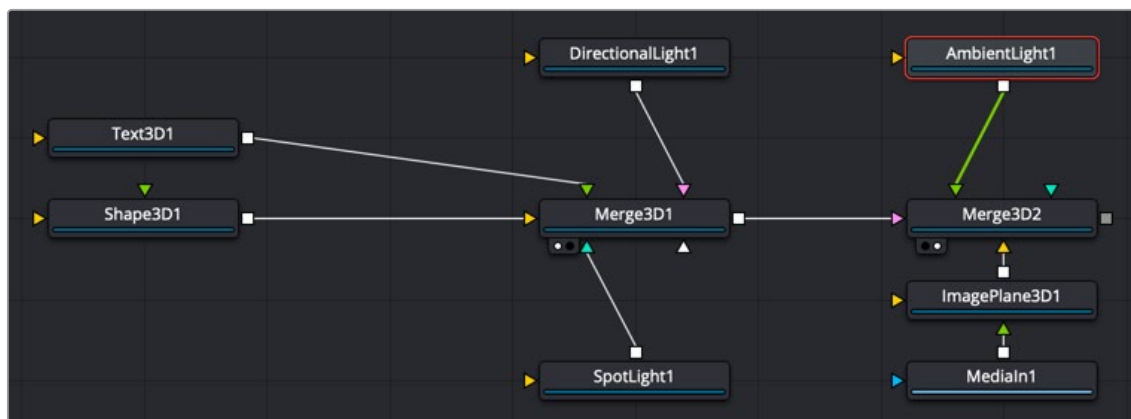
Inputs

The Ambient Light node includes a single optional orange input for a 3D scene or 3D geometry.

SceneInput: The orange input is an optional input that accepts a 3D scene. If a scene is provided, the Transform controls in this node apply to the entire scene provided.

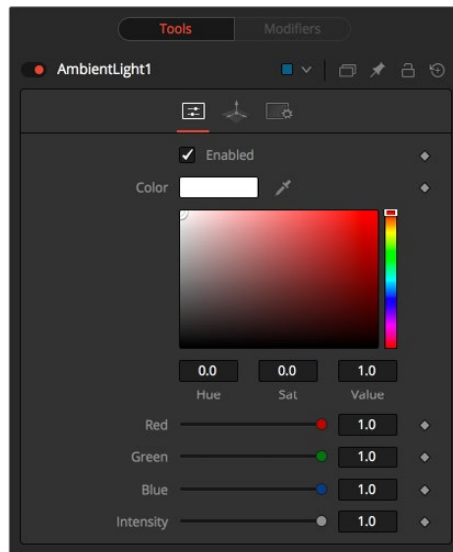
Basic Node Setup

The Ambient Light node is designed to be part of a larger 3D scene. You connect the light directly into a Merge 3D. Separating lights into different Merge 3D nodes allows you to control which lights affect which objects.



Ambient Light node structure

Inspector



Ambient Light controls

Controls Tab

The Controls tab is used to set the color and brightness of the ambient light.

Enabled

When the Enabled checkbox is turned on, the ambient light affects the scene. When the checkbox is turned off, the light is turned off. This checkbox performs the same function as the red switch to the left of the node's name in the Inspector.

Color

Use this standard Color control to set the color of the light.

Intensity

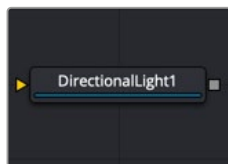
Use this slider to set the Intensity of the ambient light. A value of 0.2 indicates 20% percent light. A perfectly white texture lit only with a 0.2 ambient light would render at 20% gray (.2, .2, .2).

Common Controls

Transform and Settings Tabs

The options presented in the Transform and Settings tabs are commonly found in other lighting nodes. For more detail on the controls found in these tabs, see “The Common Controls” section at the end of this chapter.

Directional Light [3DL]



The Directional Light node

Directional Light Node Introduction

A directional light is a light with a clear direction but without a clear source or distance, similar to sunlight. This light shows an onscreen control, but the position of the control has no meaning. The rotation of the control is used to determine from where in the scene the light appears to be coming.

Similar to a Camera 3D, you connect lights into a Merge 3D and view them in the scene by viewing the Merge 3D node. Selecting a light node and loading it into the viewer does not show anything.

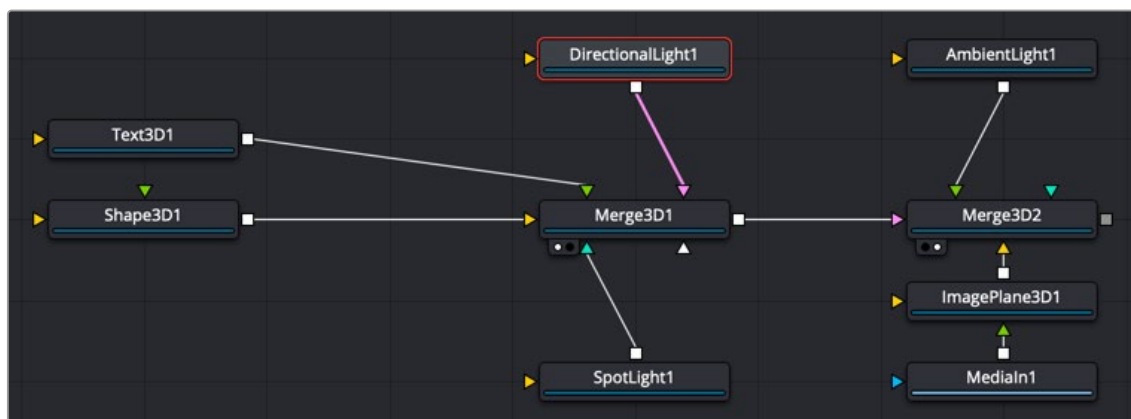
Inputs

The Directional Light node includes a single optional orange input for a 3D scene or 3D geometry.

SceneInput: The orange input is an optional input that accepts a 3D scene. If a scene is provided, the Transform controls in this node apply to the entire scene provided.

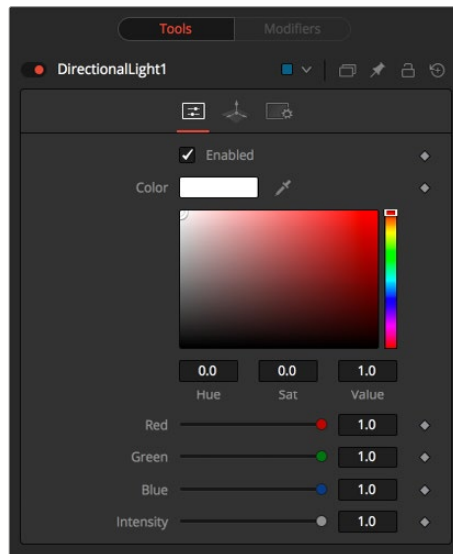
Basic Node Setup

The Directional Light node is designed to be part of a larger 3D scene. You connect the light directly into a Merge 3D. Separating lights into different Merge 3D nodes allows you to control which lights affect which objects.



Directional Light node structure

Inspector



Directional Light controls

Controls Tab

The Controls tab is used to set the color and brightness of the directional light. The direction of the light source is controlled by the rotation controls in the Transform tab.

Enabled

When the Enabled checkbox is turned on, the directional light affects the scene. When the checkbox is turned off, the light is turned off. This checkbox performs the same function as the red switch to the left of the node's name in the Inspector.

Color

Use this standard Color control to set the color of the light.

Intensity

Use this slider to set the Intensity of the directional light. A value of 0.2 indicates 20% percent light.

Shadows

This section provides several controls used to define the shadow map used when this spotlight creates shadows.

For more information, see *Chapter 85, "3D Compositing Basics,"* in the DaVinci Reference Manual or *Chapter 23* in the Fusion Reference Manual.

NOTE: The Directional light requires Hardware Renderer set as the Rendering Type in the *Renderer3D* node, for shadows to be seen in the 3D viewer.

Enable Shadows

The Enable Shadows checkbox should be selected if the light is to produce shadows. This defaults to selected.

Shadow Color

Use this standard Color control to set the color of the shadow. This defaults to black (0, 0, 0).

Density

The shadow density determines the transparency of the shadow. A density of 1.0 produces a completely opaque shadow, whereas lower values make the shadow more transparent.

Shadow Map Size

The Shadow Map Size control determines the size of the bitmap used to create the shadow map. Larger values produce more detailed shadow maps at the expense of memory and performance.

Shadow Map Proxy

Shadow Map Proxy determines the size of the shadow map used when the Proxy or Auto Proxy modes are enabled. A value of 0.5 would produce a shadow map at half the resolution defined in the Shadow Map Size.

Multiplicative/Additive Bias

Shadows are essentially textures applied to objects in the scene, so there is occasionally Z-fighting, where the portions of the object that should be receiving the shadows render over the top of the shadow. Biasing works by adding a small depth offset to move the shadow away from the surface it is shadowing, eliminating the Z-fighting. Too little bias and the objects can self-shadow themselves. Too much bias and the shadow can become separated from the surface. Adjust the Multiplicative Bias first, and then fine tune the result using the Additive Bias control.

For more information, see the *Multiplicative and Additive Bias* section of Chapter 85, “3D Compositing Basics,” in the DaVinci Resolve Reference Manual, or Chapter 25 in the Fusion Reference Manual.

Force All Materials Non-Transmissive

Normally, an RGBA shadow map is used when rendering shadows. By enabling this option, you are forcing the renderer to use a Z-only shadow map. This can lead to significantly faster shadow rendering while using a fifth as much memory. The disadvantage is that you can no longer cast “stained glass”-like shadows.

Shadow Map Sampling

Sets the quality for sampling of the shadow map.

Softness

Soft edges in shadows are produced by filtering the shadow map when it is sampled. Fusion provides two separate filtering methods for rendering shadows, which produce different effects.

NOTE: Shadows have a hard edge. No filtering of the shadow map is done at all. The advantage of this method is that you only have to sample one pixel in the shadow map, so it is fast.

- **Constant:** Shadows edges have a constant softness. A filter with a constant width is used when sampling the shadow map. Adjusting the Constant Softness slider controls the size of the filter. Note that the larger you make the filter, the longer it takes to render the shadows.
- **Variable:** The shadow edge softness grows the farther the shadow receiver is positioned from the shadow caster. The variable softness is achieved by changing the size of the filter based on the distance between the receiver and caster. When this option is selected, the Softness Falloff, Min Softness, and Max Softness sliders appear.

Constant Softness

If the Softness is set to Constant, then this slider appears. It can be used to set the overall softness of the shadow.

Softness Falloff

The Softness Falloff slider appears when the Softness is set to variable. This slider controls how fast the softness of shadow edges grows with distance. More precisely, it controls how fast the shadow map filter size grows based upon the distance between the shadow caster and receiver. Its effect is mediated by the values of the Min and Max Softness sliders.

Min Softness

The Min Softness slider appears when the Softness is set to Variable. This slider controls the Minimum Softness of the shadow. The closer the shadow is to the object casting the shadow, the sharper it is, up to the limit set by this slider.

Max Softness

The Max Softness slider appears when the Softness is set to Variable. This slider controls the Maximum Softness of the shadow. The farther the shadow is from the object casting the shadow, the softer it is, up to the limit set by this slider.

Common Controls

Transform and Settings Tabs

The options presented in the Transform and Settings tabs are commonly found in other lighting nodes. For more detail on the controls found in these tabs, see “The Common Controls” section at the end of this chapter.

DomeLight [3Do]



The Dome light

Dome Light Node Introduction

A Dome Light uses SDR and HDR images and surrounds the entire scene in a sphere light mapped with the image, similar to environment light. This light shows an onscreen control.

Inputs

This node includes an optional orange input for a 3D scene or 3D geometry, and a white Dome Texture input for connecting an image.

SceneInput: The orange input is an optional input that accepts a 3D scene. If a scene is provided, the Transform controls in this node apply to the entire scene provided.

DomeTexture: The white input connects to an SDR or HDR image.

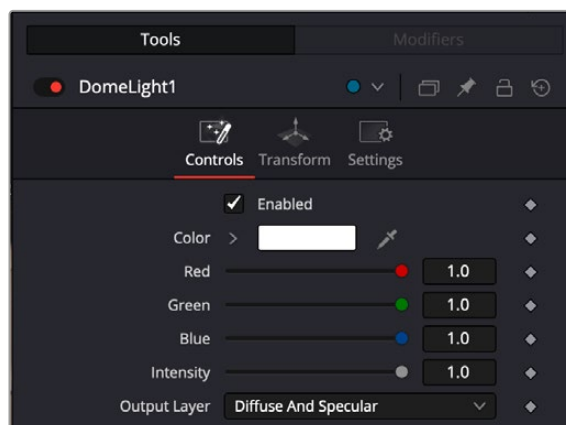
Basic Node Setup

The Dome Light node is designed to be part of a larger 3D scene. You connect the light directly into a Merge 3D. Separating lights into different Merge 3D nodes allows you to control which lights affect which objects.



Dome Light node structure

Inspector



The Dome Light controls

Controls Tab

The Controls tab is used to set the color and brightness of the uDome light. The position, direction, and scale of the light source is controlled in the Transform tab.

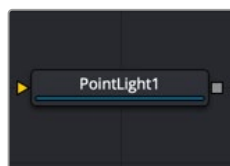
Color

Use this standard Color control to set the color of the light.

Intensity

Use this slider to set the intensity of the light. A value of 0.2 indicates 20 percent light.

Point Light [3PL]



The Point Light node

Point Light Node Introduction

A point light is a light source with a clear position in space that emits light in all directions. A light bulb is a good example of a point light.

This light shows an onscreen control, although only the position and distance of the control affect the light. Since the light is a 360-degree source, rotation has no meaning. Additionally, a point light may fall off with distance, unlike an ambient or directional light.

Similar to a Camera 3D, you connect lights into a Merge 3D and view them in the scene by viewing the Merge 3D node. Selecting a light node and loading it into the viewer does not show anything.

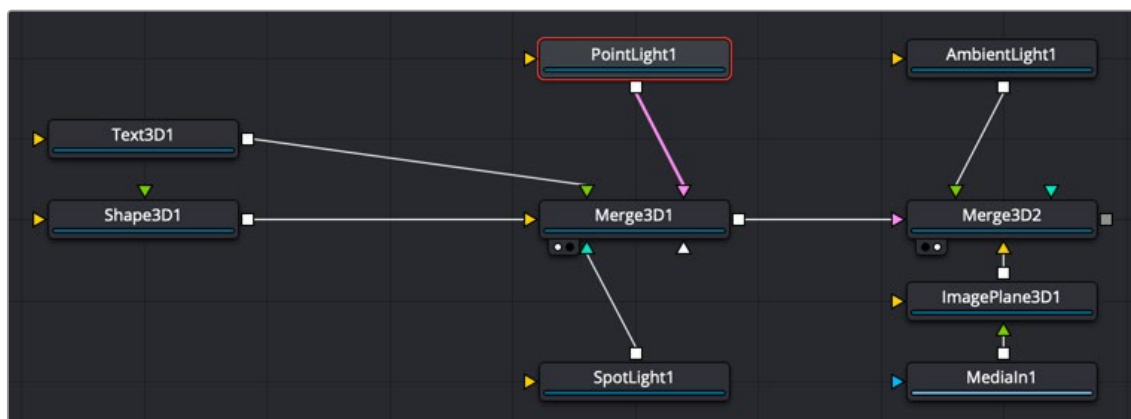
Inputs

The Point Light node includes a single optional orange input for a 3D scene or 3D geometry.

SceneInput: The orange input is an optional input that accepts a 3D scene. If a scene is provided, the Transform controls in this node apply to the entire scene provided.

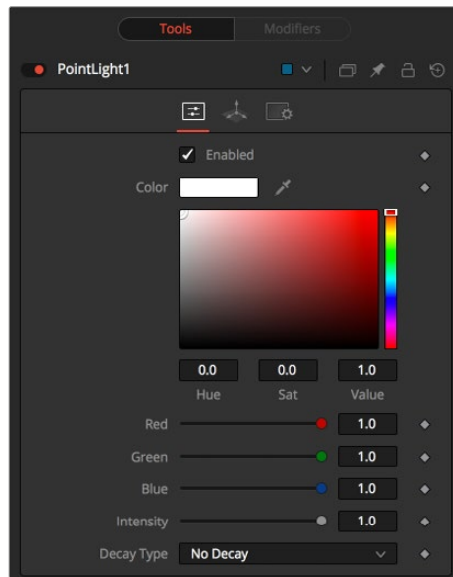
Basic Node Setup

The Point Light node is designed to be part of a larger 3D scene. You connect the light directly into a Merge 3D. Separating lights into different Merge 3D nodes allows you to control which lights affect which objects.



Point Light node structure

Inspector



Point Light controls

Controls Tab

The Controls tab is used to set the color and brightness of the point light. The position and distance of the light source are controlled in the Transform tab.

Enabled

When the Enabled checkbox is turned on, the point light affects the scene. When the checkbox is turned off the light is turned off. This checkbox performs the same function as the red switch to the left of the node's name in the Inspector.

Color

Use this standard Color control to set the color of the light.

Intensity

Use this slider to set the Intensity of the point light. A value of 0.2 indicates 20% percent light.

Decay Type

A point light defaults to No Decay, meaning that its light has equal intensity at all points in the scene. To cause the intensity to fall off with distance, set the Decay Type to either Linear or Quadratic modes.

Shadows

This section provides several controls used to define the shadow map used when this spotlight creates shadows.

For more information, see *Chapter 85, "3D Compositing Basics,"* in the DaVinci Reference Manual or Chapter 23 in the Fusion Reference Manual.

NOTE: The Point light requires Hardware Renderer set as the Rendering Type in the Renderer3D node for shadows to be seen in the 3D viewer.

Enable Shadows

The Enable Shadows checkbox should be selected if the light is to produce shadows. This defaults to selected.

Shadow Color

Use this standard Color control to set the color of the shadow. This defaults to black (0, 0, 0).

Density

The shadow density determines the transparency of the shadow. A density of 1.0 produces a completely opaque shadow, whereas lower values make the shadow more transparent.

Shadow Map Size

The Shadow Map Size control determines the size of the bitmap used to create the shadow map. Larger values produce more detailed shadow maps at the expense of memory and performance.

Shadow Map Proxy

Shadow Map Proxy determines the size of the shadow map used when the Proxy or Auto Proxy modes are enabled. A value of 0.5 would produce a shadow map at half the resolution defined in the Shadow Map Size.

Multiplicative/Additive Bias

Shadows are essentially textures applied to objects in the scene, so there is occasionally Z-fighting, where the portions of the object that should be receiving the shadows render over the top of the shadow. Biasing works by adding a small depth offset to move the shadow away from the surface it is shadowing, eliminating the Z-fighting. Too little bias and the objects can self-shadow themselves. Too much bias and the shadow can become separated from the surface. Adjust the Multiplicative Bias first, and then fine tune the result using the Additive Bias control.

For more information, see the *Multiplicative and Additive Bias* section of Chapter 85, “3D Compositing Basics,” in the DaVinci Resolve Reference Manual, or Chapter 25 in the Fusion Reference Manual.

Force All Materials Non-Transmissive

Normally, an RGBAZ shadow map is used when rendering shadows. By enabling this option, you are forcing the renderer to use a Z-only shadow map. This can lead to significantly faster shadow rendering while using a fifth as much memory. The disadvantage is that you can no longer cast “stained glass”-like shadows.

Shadow Map Sampling

Sets the quality for sampling of the shadow map.

Softness

Soft edges in shadows are produced by filtering the shadow map when it is sampled. Fusion provides two separate filtering methods for rendering shadows, which produce different effects.

- **Constant:** Shadows edges have a constant softness. A filter with a constant width is used when sampling the shadow map. Adjusting the Constant Softness slider controls the size of the filter. Note that the larger you make the filter, the longer it takes to render the shadows.
- **Variable:** The shadow edge softness grows the farther the shadow receiver is positioned from the shadow caster. The variable softness is achieved by changing the size of the filter based on the distance between the receiver and caster. When this option is selected, the Softness Falloff, Min Softness, and Max Softness sliders appear.

Constant Softness

If the Softness is set to Constant, then this slider appears. It can be used to set the overall softness of the shadow.

Softness Falloff

The Softness Falloff slider appears when the Softness is set to variable. This slider controls how fast the softness of shadow edges grows with distance. More precisely, it controls how fast the shadow map filter size grows based upon the distance between the shadow caster and receiver. Its effect is mediated by the values of the Min and Max Softness sliders.

Min Softness

The Min Softness slider appears when the Softness is set to Variable. This slider controls the Minimum Softness of the shadow. The closer the shadow is to the object casting the shadow, the sharper it is, up to the limit set by this slider.

Max Softness

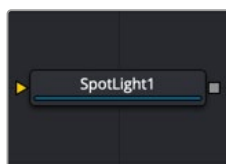
The Max Softness slider appears when the Softness is set to Variable. This slider controls the Maximum Softness of the shadow. The farther the shadow is from the object casting the shadow, the softer it is, up to the limit set by this slider.

Common Controls

Transform and Settings Tabs

The options presented in the Transform and Settings tabs are commonly found in other lighting nodes. For more detail on the controls found in these tabs, see “The Common Controls” section at the end of this chapter.

Spot Light [3SL]



The Spot Light node

Spot Light Node Introduction

A spotlight is a light that comes from a specific point and that has a clearly defined cone, with falloff of the light to the edges. Experienced stage and theatre lighting technicians may recognize the spotlight as being very similar to practical lights used in live productions. This is the only type of light capable of casting shadows.

Similar to a Camera 3D, you connect lights into a Merge 3D and view them in the scene by viewing the Merge 3D node. Selecting a light node and loading it into the viewer does not show anything.

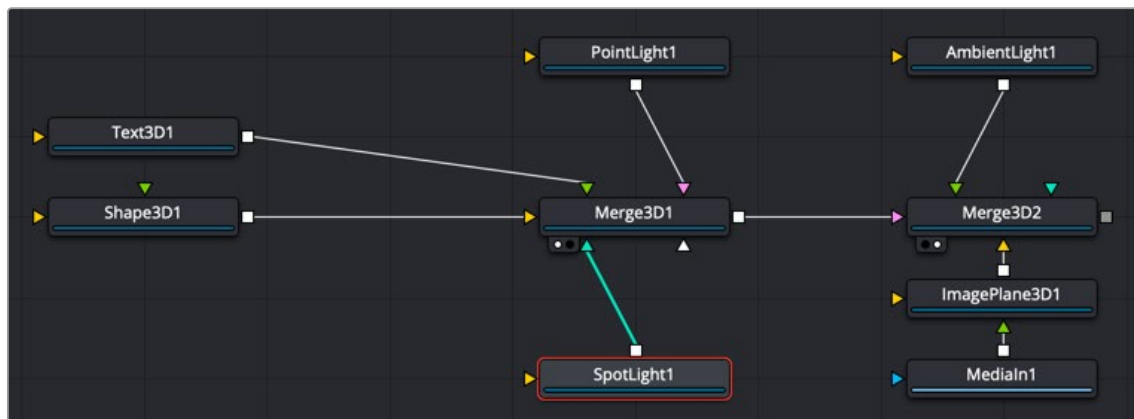
Inputs

The Spot Light node includes a single optional orange input for a 3D scene or 3D geometry.

SceneInput: The orange input is an optional input that accepts a 3D scene. If a scene is provided, the Transform controls in this node apply to the entire scene provided.

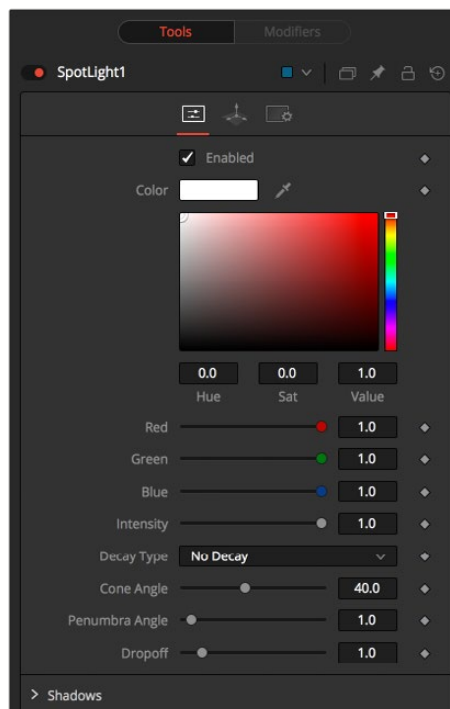
Basic Node Setup

The Spot Light node is designed to be part of a larger 3D scene. You connect the light directly into a Merge 3D. Separating lights into different Merge 3D nodes allows you to control which lights affect which objects.



Spot Light node structure

Inspector



Spot Light control

Controls Tab

The Controls tab is used to set the color and brightness of the spotlight. The position, rotation, and distance of the light source are controlled in the Transform tab.

Enabled

When the Enabled checkbox is turned on, the spotlight affects the scene. When the checkbox is turned off the light is turned off. This checkbox performs the same function as the red switch to the left of the node's name in the Inspector.

Color

Use this standard Color control to set the color of the light.

Intensity

Use this slider to set the Intensity of the spot light. A value of 0.2 indicates 20% percent light.

Decay Type

A spotlight defaults to No Falloff, meaning that its light has equal intensity on geometry despite the distance from the light to the geometry. To cause the intensity to fall off with distance, set the Decay type to either Linear or Quadratic modes.

Cone Angle

The Cone Angle of the light refers to the width of the cone where the light emits its full intensity. The larger the angle, the wider the cone angle, up to a limit of 90 degrees.

Penumbra Angle

The Penumbra Angle determines the area beyond the cone angle where the light's intensity falls off toward 0. A larger penumbra angle defines a larger falloff, while a value of 0 generates a hard-edged light.

Dropoff

The Dropoff controls how quickly the penumbra angle falls off from full intensity to 0.

Shadows

This section provides several controls used to define the shadow map used when this spotlight creates shadows.

For more information, see *Chapter 85, "3D Compositing Basics,"* in the DaVinci Resolve Reference Manual, or Chapter 25 in the Fusion Reference Manual.

Enable Shadows

The Enable Shadows checkbox should be selected if the light is to produce shadows. This defaults to selected.

Shadow Color

Use this standard Color control to set the color of the shadow. This defaults to black (0, 0, 0).

Density

The shadow density determines the transparency of the shadow. A density of 1.0 produces a completely opaque shadow, whereas lower values make the shadow more transparent.

Shadow Map Size

The Shadow Map Size control determines the size of the bitmap used to create the shadow map. Larger values produce more detailed shadow maps at the expense of memory and performance.

Shadow Map Proxy

Shadow Map Proxy determines the size of the shadow map used when the Proxy or Auto Proxy modes are enabled. A value of 0.5 would produce a shadow map at half the resolution defined in the Shadow Map Size.

Multiplicative/Additive Bias

Shadows are essentially textures applied to objects in the scene, so there is occasionally Z-fighting, where the portions of the object that should be receiving the shadows render over the top of the shadow. Biasing works by adding a small depth offset to move the shadow away from the surface it is shadowing, eliminating the Z-fighting. Too little bias and the objects can self-shadow themselves. Too much bias and the shadow can become separated from the surface. Adjust the Multiplicative Bias first, and then fine tune the result using the Additive Bias control.

For more information, see the Multiplicative and Additive Bias section of *see Chapter 85, “3D Compositing Basics,”* in the DaVinci Resolve Reference Manual, or Chapter 25 in the Fusion Reference Manual.

Force All Materials Non-Transmissive

Normally, an RGBAZ shadow map is used when rendering shadows. By enabling this option, you are forcing the renderer to use a Z-only shadow map. This can lead to significantly faster shadow rendering while using a fifth as much memory. The disadvantage is that you can no longer cast “stained glass”-like shadows.

Shadow Map Sampling

Sets the quality for sampling of the shadow map.

Softness

Soft edges in shadows are produced by filtering the shadow map when it is sampled. Fusion provides two separate filtering methods for rendering shadows, which produce different effects.

NOTE: Shadows have a hard edge. No filtering of the shadow map is done at all. The advantage of this method is that you only have to sample one pixel in the shadow map, so it is fast.

- **Constant:** Shadows edges have a constant softness. A filter with a constant width is used when sampling the shadow map. Adjusting the Constant Softness slider controls the size of the filter. Note that the larger you make the filter, the longer it takes to render the shadows.
- **Variable:** The shadow edge softness grows the further the shadow receiver is positioned from the shadow caster. The variable softness is achieved by changing the size of the filter based on the distance between the receiver and caster. When this option is selected, the Softness Falloff, Min Softness, and Max Softness sliders appear.

Constant Softness

If the Softness is set to Constant, then this slider appears. It can be used to set the overall softness of the shadow.

Softness Falloff

The Softness Falloff slider appears when the Softness is set to variable. This slider controls how fast the softness of shadow edges grows with distance. More precisely, it controls how fast the shadow map filter size grows based upon the distance between the shadow caster and receiver. Its effect is mediated by the values of the Min and Max Softness sliders.

Min Softness

The Min Softness slider appears when the Softness is set to Variable. This slider controls the Minimum Softness of the shadow. The closer the shadow is to the object casting the shadow, the sharper it is, up to the limit set by this slider.

Max Softness

The Max Softness slider appears when the Softness is set to Variable. This slider controls the Maximum Softness of the shadow. The further the shadow is from the object casting the shadow, the softer it is, up to the limit set by this slider.

Common Controls

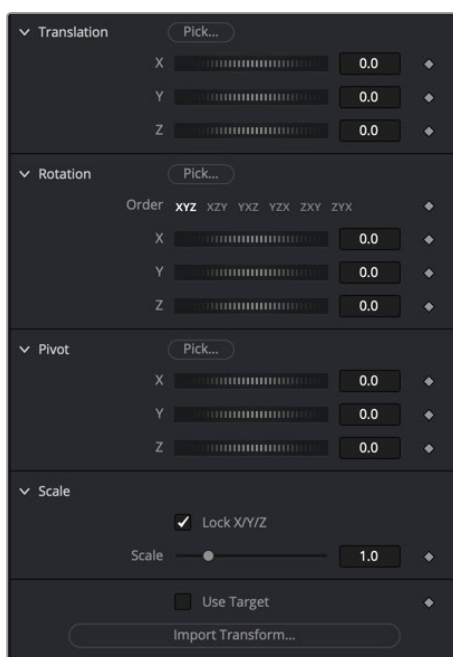
Transform and Settings Tabs

The options presented in the Transform and Settings tabs are commonly found in other lighting nodes. For more detailed information on the controls found in these tabs, see “The Common Controls” section at the end of this chapter.

The Common Controls

Nodes that handle 3D lighting share several identical controls in the Inspector. This section describes controls that are common among 3D lighting nodes.

Common Transform Tab



Common Transform 3D tab

Many tools in the 3D category include a Transform tab used to position, rotate, and scale the object in 3D space.

Translation

X, Y, Z Offset

These controls can be used to position the 3D element.

Rotation

Rotation Order

Use these buttons to select which order is used to apply Rotation along each axis of the object. For example, XYZ would apply the rotation to the X axis first, followed by the Y axis, and finally the Z axis.

X, Y, Z Rotation

Use these control to rotate the object around its pivot point. If the Use Target checkbox is selected, then the rotation is relative to the position of the target; otherwise, the global axis is used.

Pivot

X, Y, Z Pivot

A pivot point is the point around which an object rotates. Normally, an object rotates around its own center, which is considered to be a pivot of 0,0,0. These controls can be used to offset the pivot from the center.

Scale

X, Y, Z Scale

If the Lock X/Y/Z checkbox is checked, a single Scale slider is shown. This adjusts the overall size of the object. If the Lock checkbox is unchecked, individual X, Y, and Z sliders are displayed to allow individual scaling in each dimension. Note: If the Lock checkbox is checked, scaling of individual dimensions is not possible, even when dragging specific axes of the Transformation Widget in scale mode.

Use Target

Selecting the Use Target checkbox enables a set of controls for positioning an XYZ target. When Target is enabled, the object always rotates to face the target. The rotation of the object becomes relative to the target.

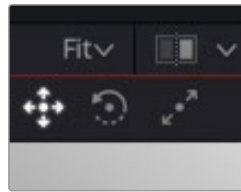
Import Transform

Opens a file browser where you can select a scene file saved or exported by your 3D application. It supports the following file types:

LightWave Scene	.lws
Max Scene	.ase
Maya Ascii Scene	.ma
dotXSI	.xsi

The Import Transform button imports only transformation data. For 3D geometry, lights, and cameras, consider using the File > FBX Import option.

Onscreen Transformation Controls



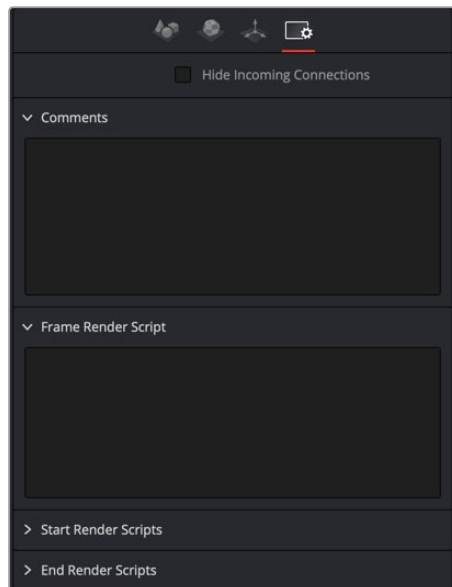
Viewer Transform buttons

Viewer Transform buttons

Most of the controls in the Transform tab are represented in the viewer with onscreen controls for transformation, rotation, and scaling. To change the mode of the onscreen controls, select one of the three buttons in the toolbar in the upper left of the viewer. The modes can also be toggled using the keyboard shortcut Q for translation, W for rotation, and E for scaling. In all three modes, individual axes of the control may be dragged to affect just that axis, or the center of the control may be dragged to affect all three axes.

The Scale sliders for most 3D tools default to locked, which causes uniform scaling of all three axes. Unlock the Lock X/Y/Z Scale checkbox to scale an object on a single axis only.

Settings Tab



Common Settings 3D controls

The Common Settings tab can be found on almost every tool found in Fusion. The following controls are specific settings for 3D nodes.

Hide Incoming Connections

Enabling this checkbox can hide connection lines from incoming nodes, making a node tree appear cleaner and easier to read. When enabled, fields for each input on a node are displayed. Dragging a connected node from the node tree into the field hides that incoming connection line as long as the node is not selected in the node tree. When the node is selected in the node tree, the line reappears.

Comment Tab

The Comment tab contains a single text control that is used to add comments and notes to the tool. When a note is added to a tool, a small red dot icon appears next to the setting's tab icon, and a text bubble appears on the node. To see the note in the Node Editor, hold the mouse pointer over the node for a moment. The contents of the Comments tab can be animated over time, if required.

Scripting Tab

The Scripting tab is present on every tool in Fusion. It contains several edit boxes used to add scripts that process when the tool is rendering. For more details on the contents of this tab, please consult the scripting documentation.